RV College of Engineering

Experiential Learning

Requirements

2025-26



Interlock: Geometric Part Identity and Contract-Validated Change Control

Team Details

Sl.No.	USN	Name
1	1RV24CS026	AKSHAT RANJAN
2	1RV24CS021	AJAY GIRISH

Mentor

Name: Prof.

Designation: Assistant Professor

Department: Computer Science **Signature**

1. Introduction

This document lists the requirements of **Interlock**, a database system for mechanical assemblies that identifies parts by their geometry, stores inter-team interfaces as machine-checkable contracts, and validates every proposed change against those contracts before it is committed. The geometry kernel is used only when files are ingested, so all queries run against ordinary indexed tables.

2. Functional Requirements

Table 2.1: Functional requirements

ID	Function	Description
FR1	Ingestion	Import STEP/BREP files headlessly; extract the occurrence tree, transforms, mass properties, cylinders and planes.
FR2	Geometric identity	Compute a pose-invariant fingerprint per solid; a re-exported, renamed or re-posed solid must map to the existing shape record.
FR3	Assembly hashing	Maintain a Merkle root hash over the assembly so two configurations compare in constant time and differences are located by descent.
FR4	Contracts	Store each revision's public face: envelope, mounting interfaces with hole patterns, mass limit, CG window, datums and attributes.
FR5	Commit validation	Validate a change as a unit, classify it as no-op, internal or breaking, and reject violations naming the constraint and affected team.
FR6	Concurrency	Accept concurrent commits without locking, using compare-and-swap on the root hash; reject only when changed paths intersect.
FR7	Tolerance chains	Evaluate declared chains by worst-case and RSS stacking against their limits; flag chains that cross team boundaries.
FR8	Structural queries	Answer explosion/BOM, mass rollup, where-used, impact, substitution and configuration-difference queries.
FR9	History	Keep an append-only commit log (author, parent and new root hash, verdict) with a full validation trace per commit.
FR10	Export	Rebuild any part or assembly as a STEP file for viewing in CAD tools such as FreeCAD.
FR11	Interfaces	Provide a command-line tool and a browser-based review interface; optionally accept natural-language questions.

3. Non-Functional Requirements

- **Performance:** structural queries must return in milliseconds and must never invoke the geometry kernel; fingerprinting is done once per solid at ingestion.
- **Correctness:** duplicate shapes must be impossible by construction; a 0.2 mm change to a hole must produce a different fingerprint.
- **Atomicity and integrity:** commits are all-or-nothing, revisions are never updated in place, and cycles in the assembly graph are rejected before insertion.
- **Storage efficiency:** storage must grow with distinct geometry rather than with the number of revisions.
- **Usability:** every rejection must state the violated constraint, the affected part and the owning team in plain language.
- **Portability and offline use:** must run on Windows, Linux and macOS without a network connection; the natural-language layer degrades to an offline router.
- **Maintainability:** modular packages (geometry, model, query, web) with an automated pytest suite.

4. Hardware Requirements

Table 4.1: Hardware requirements

Component	Minimum	Recommended
Processor	64-bit dual-core x86-64, 2.0 GHz	Quad-core x86-64, 3.0 GHz or higher
Memory (RAM)	4 GB	8 GB or more, for large STEP assemblies
Storage	2 GB free	10 GB free on an SSD
Display	Not required (kernel runs headless)	1920 × 1080, for the review interface
Network	Not required (runs fully offline)	Internet only for the optional NL layer

5. Software Requirements

Table 5.1: Software requirements

Software	Version	Purpose
Operating system	Windows 10/11, Ubuntu 22.04+, macOS 12+	64-bit host platform
Python	3.10	Implementation language
SQLite	Bundled with Python	Relational store; recursive CTE queries
cadquery-ocp (Open CASCADE)	7.9.3.1.1	Headless geometry kernel, ingestion only
NumPy, SciPy	2.2.6, 1.15.3	Invariant vectors; alignment and socket matching
TinyDB	4.9.0	Document store for non-geometric attributes
FastAPI, Uvicorn, Jinja2	0.136.3, 0.32.0, 3.1.6	Browser-based review interface
anthropic, python-dotenv	1.7.0, 1.0.1	Optional natural-language layer
pytest, httpx	9.0.3, 0.28.1	Automated testing
FreeCAD (optional)	1.1.3	Authoring models; viewing exported STEP
Web browser	Any current version	Accessing the review interface